

Data cleaning using Tableau Prep

Description: This assignment provides the opportunity to practice basic data cleaning using Tableau Prep. You will be working with a dataset that was collected through web scraping for data scientist jobs. Please answer the following question.

1. What is the data type of each column? Please categorize them using the TNOIRT schema we mentioned in class. **(Answer input to this document is needed for this question)**

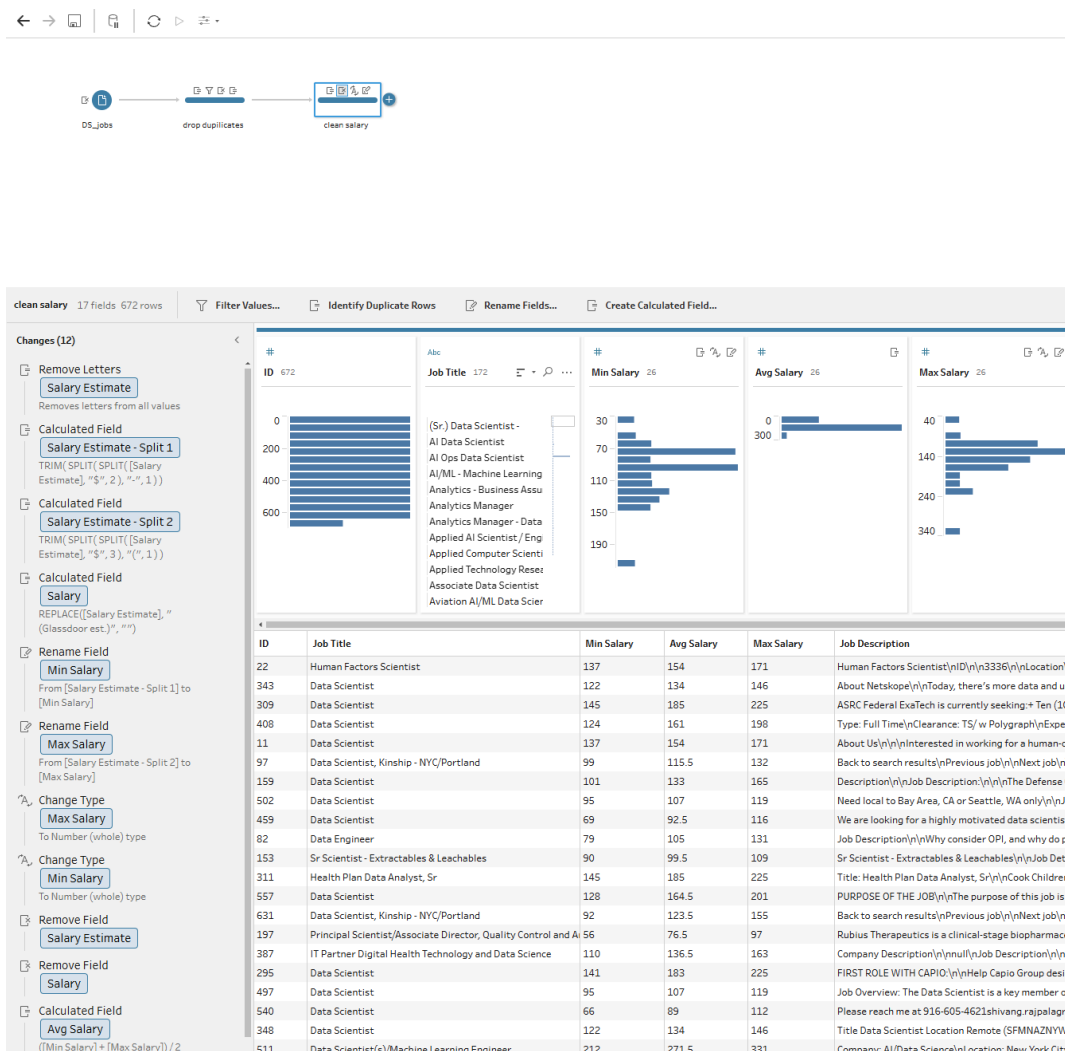
Column	ID	Job Title	Salary Estimate	Job Description	Rating	Company Name	Location	Headquarters
Type	Nominal	Textual/ Nominal	Textual	Textual	Ratio	Nominal	Nominal	Nominal
Column	Size	Founded	Type of ownership	Industry	Sector	Revenue	Competitors	/
Type	Ordinal	Temporal	Nominal	Nominal	Nominal	Ordinal	Textual	/

2. Create a new **clean step** named “**drop duplicates**” for data cleaning. There are duplicated records in the dataset (ID identifies each unique response). Please remove the duplicates. How many unique records are left?

To ensure data quality, I identified duplicate rows using the “Is Duplicate Row?” function in Tableau Prep. I then applied a filter to exclude all rows flagged as Duplicate, keeping only unique records. Before: 679 rows. After: 672 rows.

3. Create a new **clean step** named “**clean salary**”. Salary Estimate column has the salary level for each response. Most of the values (not all) are expressed with a redundant suffix “(Glassdoor est.)”. Please create a new column “**Salary**” that unifies them without the suffix. Describe your operations with screenshot(s).

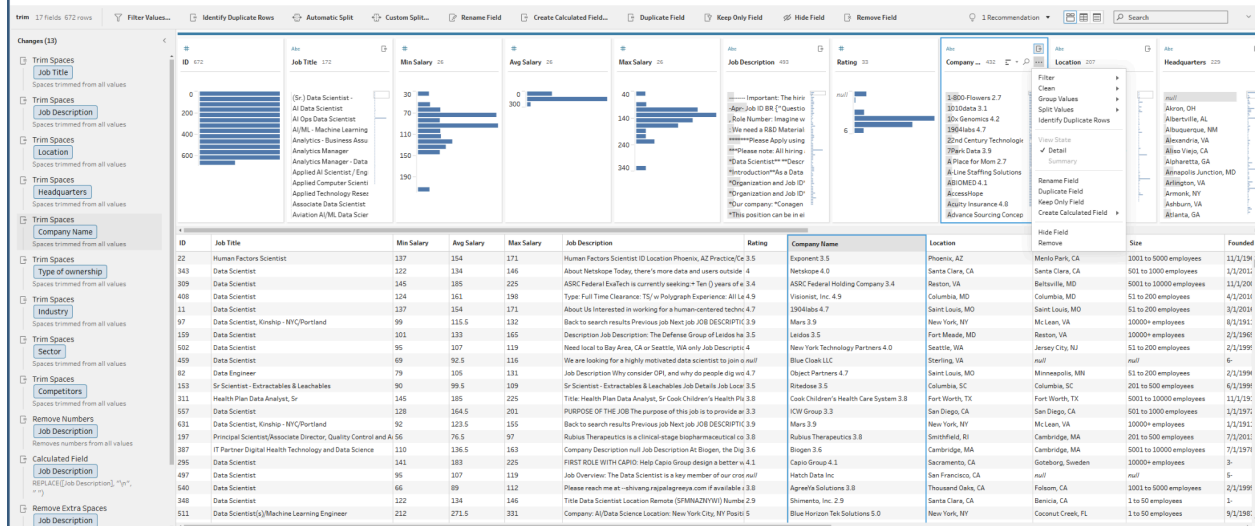
I created min, max, and avg salary columns.



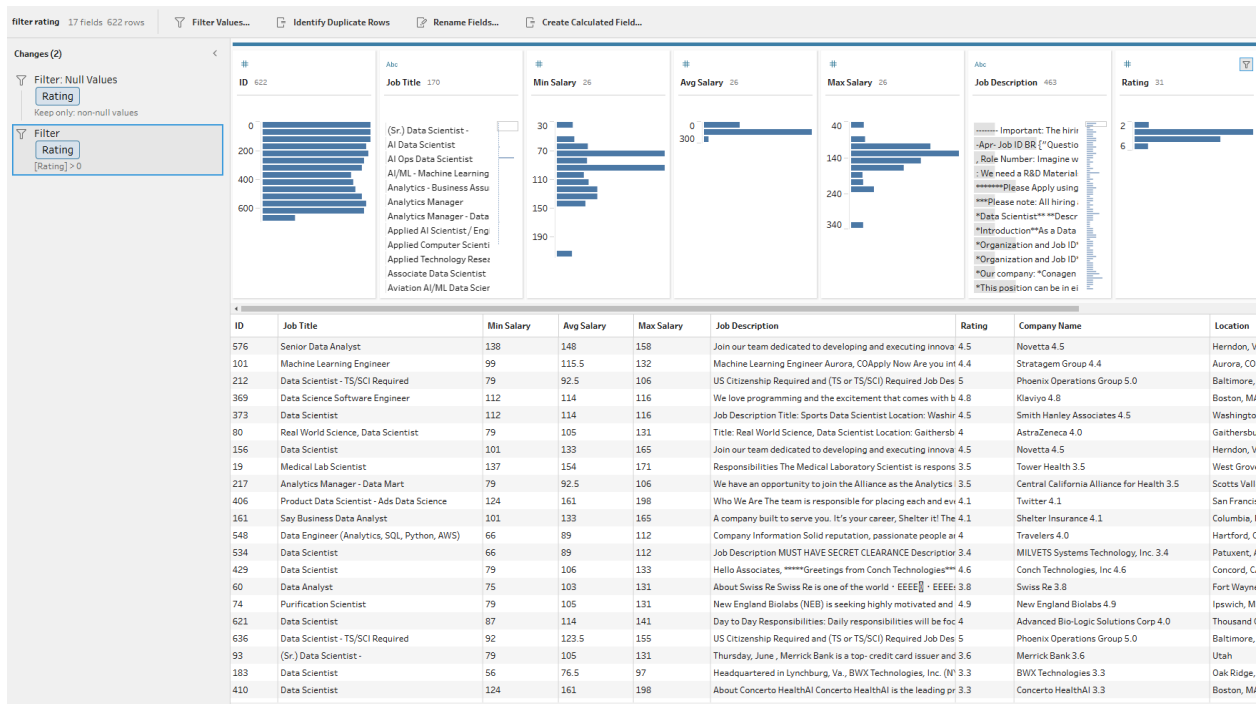
4. Create a new **clean step** named **"trim"**. Values in some columns that have white space before, or after the value. Please remove the extra space with cells of each **TEXT** field. Describe your operations with screenshot(s).

In this step, I focused on cleaning up extra white space from text fields. Tableau Prep's Trim Spaces option was applied across text columns, including Job Title, Job Description, Company Name, Location, Headquarters, Type of Ownership, Industry, Sector, and Competitors. This removed any leading or trailing white spaces that could interfere with consistency. I also cleared

spaces and numbers from the text columns using clean.



5. Create a new **clean** step named **“filter rating”**. For Rating, some values are 0, while others left the value empty in order to indicate no ratings available. These ratings are invalid. Please filter all rows with empty ratings (null) and keep rows with non-empty ratings. How many rows are left? Describe your operations with screenshot(s).



In this step, I cleaned the Rating column by removing invalid values. The dataset contained both null values (blank ratings) and 0 values, which do not represent actual ratings. To fix this, I applied two filters in Tableau Prep: Filter Null Values Exclude Null Values on the Rating field. A calculation filter to remove

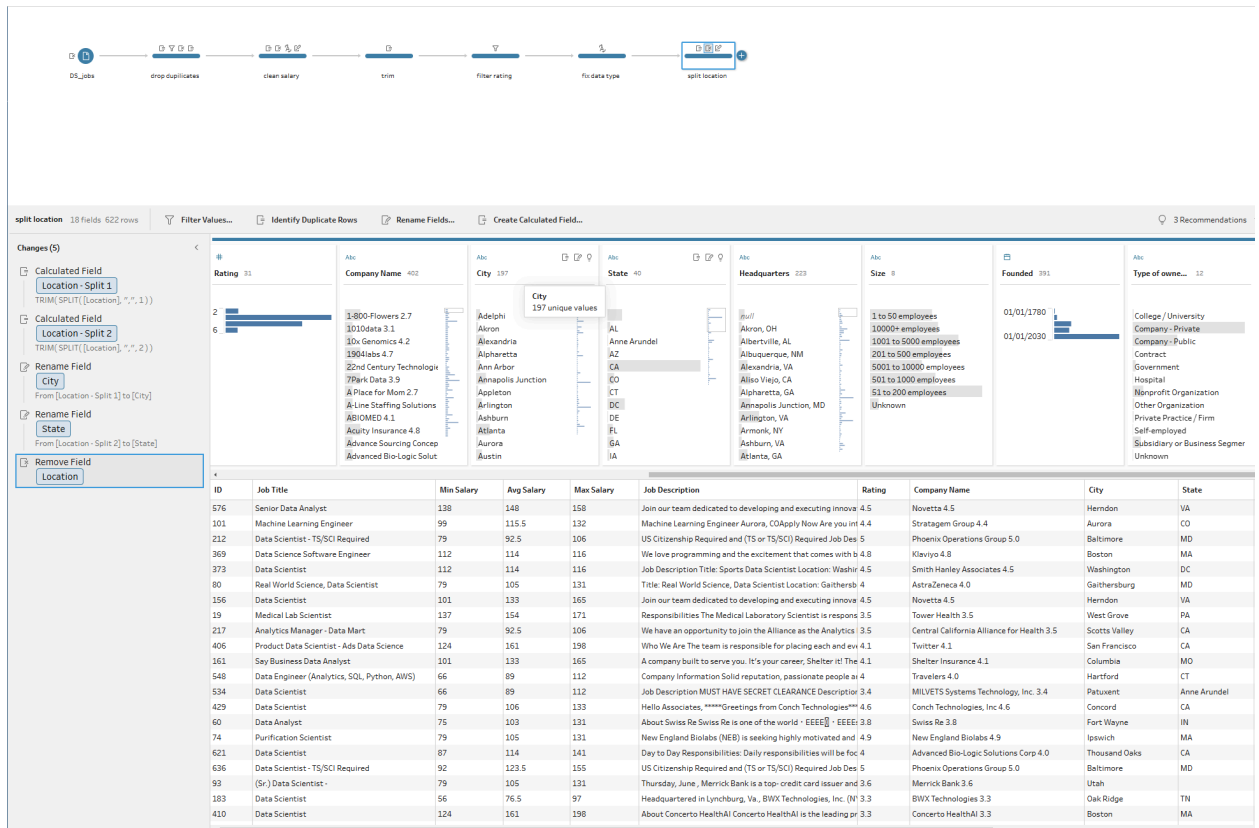
rows where the rating was equal to 0. After applying these filters, the dataset was reduced from 672 rows to 622 rows, leaving only entries with valid, non-empty ratings.

- Create a new **clean step** named “**fix data type**”. Examine the column "Founded". It is currently assigned to a String data type (indicated by “Abc”). Please convert it to the date data type. Describe your operations with screenshot(s).

Min Salary	Avg Salary	Max Salary	Job Description	Rating	Company Name	Location	Headquarters	Size	Founded	Type of ownership	Industry
138	148	158	Join our team dedicated to developing and executing innova	4.5	Nuvetta 4.5	Herndon, VA	McLean, VA	501 to 1000 employees	07/01/2012	Company - Private	Health
99	115.5	132	Machine Learning Engineer Aurora, COApply Now Are you int	4.4	Stratagem Group 4.4	Aurora, CO	Aurora, CO	1 to 50 employees	02/01/2007	Company - Private	Accounting
79	92.5	106	US Citizenship Required and (TS or TS/SC) Required Job Des	5	Phoenix Operations Group 5.0	Baltimore, MD	Woodbine, MD	1 to 50 employees	03/01/2011	Company - Private	Advertising & Marketing
112	114	116	We love programming and the excitement that comes with it	4.8	Klawns 4.8	Boston, MA	Boston, MA	201 to 500 employees	10/01/2012	Company - Private	Aerospace & Defense
112	114	116	Job Description Title: Sports Data Scientist Location: Washi	4.5	Smith Hanley Associates 4.5	Washington, DC	New York, NY	1 to 50 employees	07/01/1980	Company - Private	Architectural & Engineer
79	105	131	Title: Real World Science, Data Scientist Location: Gatherb	4	AstraZeneca 4.0	Cambridge, UK	Cambridge, UK	10000+ employees	01/01/2013	Company - Public	Hospital
101	133	165	Join our team dedicated to developing and executing innova	4.5	Nuvetta 4.5	Herndon, VA	McLean, VA	501 to 1000 employees	09/01/2012	Company - Private	Biotech & Pharmaceutical
137	154	171	Responsibilities The Medical Laboratory Scientist is respons	3.5	Tower Health 3.5	West Grove, PA	Reading, PA	5001 to 10000 employees	09/01/2017	Nonprofit Organization	Cable, Internet & Teleph
79	92.5	106	We have an opportunity to join the Alliance as the Analytica	3.5	Central California Alliance for Health 3.5	Scotts Valley, CA	Scotts Valley, CA	501 to 1000 employees	09/01/1996	Nonprofit Organization	Colleges & Universities
124	161	198	Who We Are The team is responsible for placing each and eve	4.1	Talman 4.1	San Francisco, CA	San Francisco, CA	1001 to 5000 employees	05/01/2006	Company - Public	Chemical Manufacturing
101	133	165	A company built to serve you. It's your career. Shelter-HI The	4.2	Shelter Insurance 4.2	Columbia, MD	Columbia, MD	1001 to 5000 employees	01/01/1946	Company - Private	Computer Hardware & S
66	89	112	Company Information Solid reputation, passionate people at	4	Travelers 4.0	Hartford, CT	Hartford, CT	10000+ employees	10/01/1853	Company - Public	Unknown
66	89	112	Job Description MUST HAVE SECRET CLEARANCE Descriptio	3.4	MLVETS Systems Technology, Inc. 3.4	Patterson, CA	Orlando, FL	51 to 200 employees	01/01/1986	Company - Private	
79	106	133	Hello Associates, ""Greetings from Conch Technologies""	4.6	Conch Technologies, Inc 4.6	Concord, CA	Memphis, TN	51 to 200 employees	02/01/1966	Company - Private	
75	103	131	About Swiss Re Swiss Re is one of the world - EEE EEE EEE	3.8	Swiss Re 3.8	Fort Wayne, IN	Zurich, Switzerland	10000+ employees	07/01/1863	Company - Public	
79	105	131	New England Biolabs (NEB) is seeking highly motivated and	4.9	New England Biolabs 4.9	Ipswich, MA	Ipswich, MA	201 to 500 employees	06/01/1974	Company - Private	
87	114	141	Day to Day Responsibilities: Daily responsibilities will be fac	4	Advanced Bio-Logic Solutions Corp 4.0	Thousand Oaks, CA	Westlake Village, CA	1 to 50 employees	02/01/2000	Company - Private	
92	123.5	155	US Citizenship Required and (TS or TS/SC) Required Job Des	5	Phoenix Operations Group 5.0	Baltimore, MD	Woodbine, MD	1 to 50 employees	08/01/2011	Company - Private	
79	105	131	Thursday, June - Merrick Bank is a top-credit card issuer and	3.6	Merrick Bank 3.6	Utah	South Jordan, UT	201 to 500 employees	06/01/1997	Company - Private	
56	76.5	97	Headquartered in Lynchburg, Va., BVX Technologies, Inc. (N	3.3	BVX Technologies 3.3	Oak Ridge, TN	Lynchburg, VA	5001 to 10000 employees	07/01/1850	Company - Public	
124	161	198	About Concerto HealthAI Concerto HealthAI is the leading pr	3.3	Concerto HealthAI 3.3	Boston, MA	Boston, MA	201 to 500 employees	12/01/2018	Company - Private	

In this step, I corrected the data type of the Founded column.

- Create a new **clean step** named “**split location**”. The Location column collects the city information for each response. However, they put the state name with the city names. We can see that city name and state name seem to be separated by a comma. Please split the column by comma and name the two split columns “city” and “state”. Describe your operations with screenshot(s).



Split location column into City and State.

8. Your data is looking good as follows and you're ready to generate your output file.

Tableau Prep Builder - Flow1*

File Edit Flow Server Help

Connections

DS_jobs.xlsx
Microsoft Excel

split location 19 fields 625 rows

Filter Values... Identify Duplicate Rows

3 Recommendations

Search

Tables

Use Data Interpreter
Data Interpreter might be able to clean your Microsoft Excel workbook.

DS_jobs

Flow diagram showing steps: DS_jobs, drop duplicates, clean salary, trim, filter rating, fix data type, split location.

split location 19 fields 625 rows

Rating 32

Company Name 405

Location 199

state 44

city 197

Headquarters 225

Size

Is Duplicate Row?	ID	Job Title	Salary Estimate	Salary	Job Description	Rating	Company Name
Unique	183	Data Scientist	\$56K-\$97K (Glassdoor est.)	\$56K-\$97K	Headquartered in Lynchburg, Va., BWX Technologies, Inc.	3.3	BWX Technologies
Unique	485	Data Scientist	\$31K-\$56K (Glassdoor est.)	\$31K-\$56K	Join us in making roads safer by making drivers better!	4.9	Cambridge Mobile
Unique	531	Data Scientist	\$66K-\$112K (Glassdoor est.)	\$66K-\$112K	Job Description: Data Scientist - EEEEEE - EEEEEE Washi	5	JCD Staffing
Unique	245	Data Scientist	\$90K-\$124K (Glassdoor est.)	\$90K-\$124K	Job Overview: Advanced Analytics - Data Scientist v	3.7	Sanofi
Incomplete	423	Data Scientist	\$124K-\$198K (Glassdoor est.)	\$124K-\$198K	Who we are: Criteo (NASDAQ: CRI) is the global tech	3.9	Criteo

Create a new **output step** named “**split location**”. Save your processed data as an Excel file and name it **Output.xlsx**. You may use any worksheet name.

Save your **Tableau Flow (File -> Save)** as **Flow1.tfl**. You are ready to submit the processed data and Tableau flow!